Pal 2021 Breast Cancer — analysis summary

RUN 2 OF 3

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Batch mixing + continuity, judged on the regenerated embedding across a θ × n_pcs sweep (per-neighbourhood iLISI). At θ=10/n_pcs=20 mixing plateaus (median iLISI 4.96 of ~22); every UMAP band holds 18–19 samples on one connected manifold (8 island cells removed).

Gate 2 — module geography → GO (non-branching)

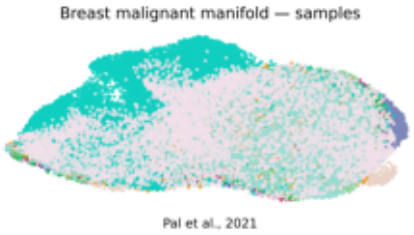
One organized developmental axis: pSom high at the root, Crest high at the terminus (Crest↔pSom spatial r=−0.93; Gast↔pSom r=−0.57), an inverse terminal Crest gradient → GO. FLAG: Crest co-enriches the Gast side (r=+0.40 vs pSom −0.93), an inversion.

Gate 3 — hub cross-repression → NA

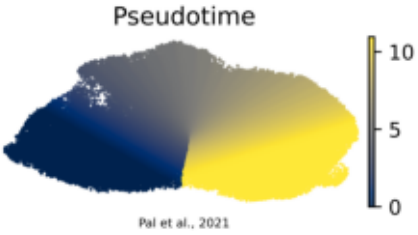
Branching-only. The manifold is non-branching (PPT forks scale with node count 10→21; a curve fits fold-free, 2 tips / 0 forks), so there is no bifurcation to test for Gast↔pSom cross-repression.

STEP FIGURES

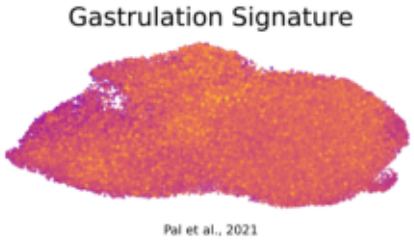
Manifold — samples



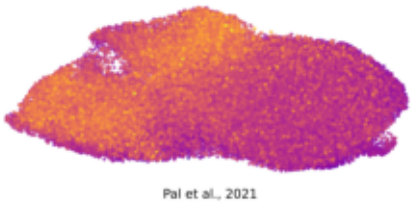
Pseudotime



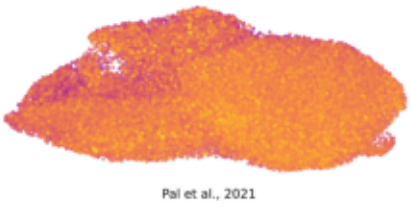
Gastrulation



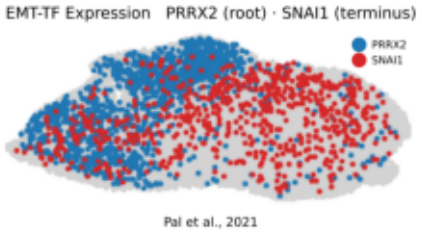
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Pal et al., 2021), 10x Chromium — 59,359 malignant cells · 22 samples (ER+, HER2+, TNBC).

Verdict.
The embryonic pattern reproduces as a non-branching axis (Gate 1 GO / Gate 2 GO / Gate 3 NA): a single gastrulation/somitogenesis origin resolves toward a neural-crest terminus, with no Gast/pSom bifurcation. Crest-arm inversion flagged. Module strengths Gast −0.007 / pSom +0.036 / Crest −0.034 (Linear-pSom/Crest). EMT-TFs PRRX2 (early/root) ↔ SNAI1 (late/terminus).

Parameter choices

Sample threshold	≥12 cells (geom mean of 56→3 gap); 22/26 samples kept	**Integration**	Harmony θ=10 (iLISI plateau 5→10), n_pcs=20
Overdispersed genes	3,733 (scFates find_overdispersed)	**Island removal**	radius graph 6× median-NN, largest component (8 cells dropped)
Geometry	Non-branching (ElPiGraph curve; 2 tips, 0 forks, fold-free SA=2.52)	**Root**	pSom/Gast-high, Crest-low tip (origin; cyc ρ=−0.57)
Module means	Gast −0.007 / pSom +0.036 / Crest −0.034 (Linear-pSom/Crest)	**EMT-TFs**	PRRX2 (early/root) ↔ SNAI1 (late/terminus)